Task 8 – Configure Continuous Deployment

Student: Prince Raj

Objective

Configure continuous deployment using Jenkins and Kubernetes by automating the application of Kubernetes Deployment and Service manifests and verifying the deployed resources.

Jenkins Continuous Deployment Pipeline

A Jenkins Pipeline named **kubernetes-continuous-deployment** was configured to execute kubectl commands against the Docker Desktop Kubernetes cluster.

Successful Build

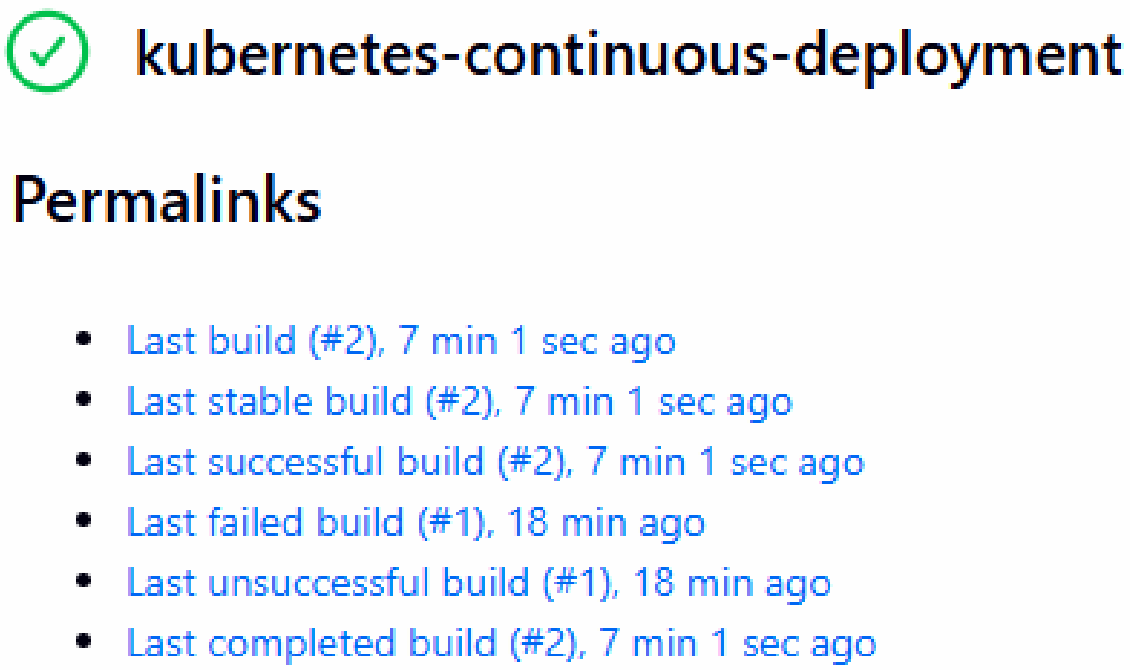


Figure 1. Jenkins Build #2 completed successfully.

Deployment Verification – Jenkins Console Output

The successful Jenkins build verified the Kubernetes node, applied the deployment.yaml and service.yaml manifests, checked rollout status, and verified the Pod and NodePort Service.

```
Started by user Anand Kumar
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\kubernetes-continuous-deployment
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Verify Kubernetes)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\kubernetes-continuous-deployment>"C:/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin/kubectl.exe" get nodes
NAME                   STATUS   ROLES           AGE    VERSION
desktop-control-plane   Ready    control-plane   103m   v1.36.1
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy Application)
[Pipeline] dir
Running in C:/Users/HP/kubernetes-task-1
[Pipeline] {
[Pipeline] bat

C:\Users\HP\kubernetes-task-1>"C:/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin/kubectl.exe" apply -f deployment.yaml
deployment.apps/nginx-demo configured
[Pipeline] bat

C:\Users\HP\kubernetes-task-1>"C:/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin/kubectl.exe" apply -f service.yaml
service/nginx-demo-service unchanged
[Pipeline] }
[Pipeline] // dir
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Deployment)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\kubernetes-continuous-deployment>"C:/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin/kubectl.exe" rollout status deployment/nginx-demo
deployment "nginx-demo" successfully rolled out
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Pods)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\kubernetes-continuous-deployment>"C:/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin/kubectl.exe" get pods -o wide
NAME                          READY   STATUS    RESTARTS   AGE   IP           NODE                   NOMINATED NODE   READINESS GATES
nginx-demo-6f5d9c55b8-pp2nc   1/1     Running   0          38m   10.244.0.11   desktop-control-plane   <none>           <none>
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Service)
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\kubernetes-continuous-deployment>"C:/Users/HP/AppData/Local/Programs/DockerDesktop/resources/bin/kubectl.exe" get service nginx-demo-service
NAME                TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
nginx-demo-service   NodePort   10.96.168.87   <none>        80:31450/TCP   63m
```

Figure 2. Jenkins console showing successful Kubernetes deployment and verification.

Result

Continuous deployment was successfully demonstrated using Jenkins and Kubernetes. The pipeline automatically applied the Kubernetes manifests and verified the running Deployment, Pod, and NodePort Service.